

DOCKHUB · DESIGN DECISIONS REGISTER

DockHub design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

DockHub design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction	Accept the changes P1 to P12 as made, or ask for changes	Accept	The whole build plan	DKH-DDR-003, Table 1
2	Bay door hinges on the outside of the front panel (R10)	(a) stainless piano hinges with peened pins, as modelled; (b) concealed hinges, needing a wider bay pitch or welded hinges	(a) for the prototype	Bay doors (build plan section 3.15, step 17)	DKH-DDR-003, A2
3	How to close R6 (9.3 % solar against 10 %)	Accept about 9 %; route flat packs to bay 1 in daylight and accept occasional waits; feed more bays from the sun; drop the target	None yet	Controller logic and MPPT wiring; not the cabinet build	DKH-DDR-002, Table 2
4	How to close R9 (charging pauses above about 29 to 39 °C ambient)	Accept daytime pauses in hot climates; add active cooling; narrow the design ambient; raise the charge limit with SwapCell	None yet	Fans and vent hood if cooling is added	DKH-DDR-002, Table 2
5	Pack ownership model	Station-owned pool, rider-owned one-for-one exchange, fleet packs	None yet; to follow co-design	Not part of the build	DKH-DDR-001 item 10
6	First pilot site and partner	A delivery-worker group in a high-income city, or a moto-taxi hub in East Africa	None yet	Siting, supply and anchoring (safety stop S7)	DKH-DDR-001 item 11
7	Propagation test and standard for R8	Which lithium-ion propagation test a later version aims for, and whether a workshop-built cabinet can pass it	None yet	Liners, plenum and fire unit at TRL 4 and later	DKH-PRC-001, Open questions
8	Fire suppression	Keep the condensed-aerosol unit, or use sand or vermiculite-filled bay floors and a sealed, vented compartment	None yet	Fire unit (build plan step 5)	DKH-PRC-001, Open questions
9	Siting permissions and electrical supply host in the first pilot city	To be found with the pilot partner	None yet	Safety stops S2 and S7	DKH-PRC-001, Open questions
10	Riders who do not want to share identity	Anonymous prepaid cards or another token	None yet	Access panel software only	DKH-PRC-001, Open questions
11	Product renders: clear windows in the bay doors	Render-only windows with solid steel doors in the build, or fire-rated glazing priced against R16	Solid steel doors in the build; windows render-only	None (the build plan has solid doors)	Review note, 2026-09-26, item 1

#	Decision needed	Options	Recommendation	Affects in the build	Source
12	Product renders: scene state, header sign and markings, four-bay fit caption	Accept as render choices	Accept; settle sign wording with branding	None	Review note, 2026-09-26, items 2, 3 and 5
13	Update the photoreal renders, card and social preview to the constructable design	Re-render on Amish's Mac after decision 1	Re-render after decision 1	None	DKH-DDR-003, Consequences

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Charger footprint and fixing holes, and that its power factor is 0.9 or better	The shelf holes are marked from the charger; R5 holds only with power-factor-corrected chargers	DKH-CAL-001 section 6; BOM line 5
2	Controller, MPPT and grid box sizes (no larger than 220 x 100 x 190, 150 x 70 x 180 and 220 x 92 x 200 mm)	They must fit between the floor flange and the shelf flange on the back panel	DKH-DDR-003, P9
3	Solenoid lock size (about 30 x 30 x 50 mm) and bolt throw	The tongue position on the door is set from it	DKH-DDR-003, P6
4	SwapCell v0.3 receptacle flange and fixing	The cradle's 72 x 50 x 9 mm pocket and its two M3 inserts are set from it	DKH-DDR-003, P7
5	Fans 120 x 120 x 25 mm with corner holes on a 105 mm square	The roof holes and spacer positions are set from them	DKH-DDR-003, P5
6	Aerosol unit size (about 76 x 420 mm) and its straps	Its position under the roof line, clear of the fans and beams	DKH-DDR-003, P9
7	Solar panel size, frame depth (35 mm) and that the maker allows end clamps	The rail length and clamps are set from it	DKH-DDR-003, P3
8	A printable flame-retardant filament rated UL 94 V-0	The cradle material	DKH-DDR-003, P7
9	Anchors with a design tension of at least 3.8 kN in the host concrete, and the pad	R15 stays at risk until they are chosen; not in the BOM	DKH-CAL-001 section 11

■ Value engineering

Value-engineering target: USD 1,200 (a hypothetical control target, not a limit), for the four-bay cabinet with two bays fitted and SwapCell packs excluded. Estimated cost of the constructable design: USD 1,214 (USD 14 over the target). All four bays fitted cost an estimated USD 1,438; each further bay adds USD 112.

Main cost drivers: the parts added to make the design buildable took the estimate from USD 1,199 to USD 1,214, namely the canopy end clamps, post plates and bolts (line 12, USD 60 to USD 70) and the service door hinge (line 15, USD 30 to USD 35). All prices are indicative.

Savings worth trying: recovering the USD 14 at supplier selection.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items with a recommendation: four-bay cabinet with bays 1 and 2 fitted and a \$1,200 value-engineering target (budget_usd) for that fit; 400 W canopy kept and R6 relaxed to 10 %; four bays in one row; one charger and CAN channel per bay; steel compartment per bay with a rear plenum; doors locked during a fire alarm; MPPT to bay 1; offline-first NFC access; 70 % health gate	Amish: "i accept all your recommendations, go with them across all repos."	DKH-DDR-001, DKH-DDR-002
2026-09-25	TRL 3 engineering proposals: build to SwapCell interface v0.3; release packs at 80 % state of charge or more; chargers with a power factor of 0.9 or better; plinth flush with the body; 15 min rider patience and the demand profile of DKH-CAL-001; two-bay first prototype that cannot run the duty case	Amish, same instruction	DKH-DDR-001 items 12 and 13, DKH-DDR-002
2026-09-30	Make each design physically buildable while drawing the build plan, recording every change	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The resulting changes are open (decision 1)	DKH-DDR-003
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	STANDARDS section 18